

Fall 2000

Problem 1 (Fa00). Let V be a finite dimensional vector space, and let $f : V \rightarrow V$ be a linear transformation. Let W denote the image of f . Prove that the restriction of f to W , considered as an endomorphism of W , has the same trace as $f : V \rightarrow V$.

Problem 2 (Fa00). Let K be a subset of a compact metric space (X, d) . Assume that, for every continuous function $f : X \rightarrow \mathbb{R}$, the restriction of f to K attains a maximum on K . Prove that K is compact.

Note: See also ??.

Problem 3 (Fa00). Suppose $\mathbb{K}$ is a field and R is a nonzero $\mathbb{K}$ -algebra generated by two elements a and b which satisfy $a^2 = b^2 = 0$ and $(a + b)^2 = 1$. Show R is isomorphic to $M_2(\mathbb{K})$, the algebra of 2×2 matrices over $\mathbb{K}$.

Problem 4 (Fa00). Evaluate the integral

$$I = \frac{1}{2\pi i} \int_{|z|=1} \frac{dz}{\sin 4z}$$

where the direction of integration is counterclockwise.

Problem 5 (Fa00). Let f be a real valued differentiable function on $(-1, 1)$ such that $f(x)/x^2$ has a finite limit as $x \rightarrow 0$. Does it follow that $f''(0)$ exists? Give a proof or a counterexample.

Problem 6 (Fa00). Suppose V is a vector space over a field $\mathbb{K}$. If U and W are subspaces, let $E(U, W)$ be the set of linear endomorphisms f of V over $\mathbb{K}$ with the property that the image of $f(U)$ in V/W is finite dimensional. Show that $E(U, U)$ is a subring of the ring of endomorphisms of V with two-sided ideals $E(V, U)$ and $E(U, 0)$.

Problem 7 (Fa00). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be uniformly continuous with $f(0) = 0$. Prove: there exists a positive number B such that $|f(x)| \leq 1 + B|x|$, for all x .

Problem 8 (Fa00). Let $\mathbb{F}_p$ denote the field of p elements, p a prime. Let n be a positive integer. Prove that there is a transformation $A \in \text{GL}_n(\mathbb{F}_p)$, the group of invertible linear transformations from $\mathbb{F}_p^n$ into itself, which, as a permutation of the nonzero vectors of $\mathbb{F}_p^n$, acts as a single cycle of length $p^n - 1$.

Problem 9 (Fa00). Assume the nonconstant entire function f takes real values on two intersecting lines in the complex plane. Prove that the measure of either angle formed by the lines is a rational multiple of π .

Problem 10 (Fa00). Find all real numbers t for which the quadratic form Q_t on $\mathbb{R}^3$, defined by

$$Q_t(x_1, x_2, x_3) = 2x_1^2 + x_2^2 + 3x_3^2 + 2tx_1x_2 + 2x_1x_3$$

is positive definite.

Problem 11 (Fa00). Let $f_n : \mathbb{R}^k \rightarrow \mathbb{R}^m$ be continuous ($n = 1, 2, \dots$). Let K be a compact subset of $\mathbb{R}^k$. Suppose $f_n \rightarrow f$ uniformly on K . Prove that

$$S = f(K) \cup \bigcup_{n=1}^{\infty} f_n(K)$$

is compact.

Problem 12 (Fa00). Show that for each positive integer k there exists a positive integer N such that there are at least k nonisomorphic groups of order N .

Problem 13 (Fa00). Let $f(z)$ be the rational function $p(z)/q(z)$, where $p(z)$ and $q(z)$ are nonzero polynomials with complex coefficients, such that the degree of $p(z)$ is less than the degree of $q(z)$, and such that $q(z)$ has no complex zeros with nonnegative imaginary part. Prove that if z_0 is a complex number with positive imaginary part, then

$$f(z_0) = \frac{1}{2\pi i} \int_{-\infty}^{+\infty} \frac{f(t)}{t - z_0} dt$$

Problem 14 (Fa00). Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear transformation, where $n > 1$. Prove that there is a 2-dimensional subspace $M \subseteq \mathbb{R}^n$ such that $T(M) \subseteq M$.

Problem 15 (Fa00). Suppose that $f : \mathbb{R} \rightarrow \mathbb{R}$ is a nonconstant function such that $f(x) \leq f(y)$ whenever $x \leq y$. Prove that there exist $a \in \mathbb{R}$ and $c > 0$ such that $f(a + x) - f(a - x) \geq cx$ for all $x \in [0, 1]$.

Problem 16 (Fa00). Let G be a finite group of order n with the property that for each divisor d of n there is at most one subgroup in G of order d . Show that G is cyclic.

Problem 17 (Fa00, Fa24). Let U be a connected and simply connected open subset of the complex plane, and let f be a holomorphic function on U . Suppose a is a point of U such that the Taylor series of f at a converges on an open disc D that intersects the complement of U . Does it follow that f extends to a holomorphic function on $U \cup D$? Give a proof or a counterexample.

Problem 18 (Fa00). Let R be a ring with identity, having fewer than eight elements. Prove that R is commutative.